

EDA TOOLS - Synopsys, cadence

APR EDA runs FUSION COMPILER, ICC2, innovus, encounter (its older version of innovus).

Floorplan – innovus (some product-based companies will prefer innovus because if we have 500 -600 macros it will be easy to place those)

PNR both tools

get_cells/get_net/ get_pins (These commands are same for both the tools)

sdc commands both tool can use.

UPF(Synopsys) and CPE(cadence) are same

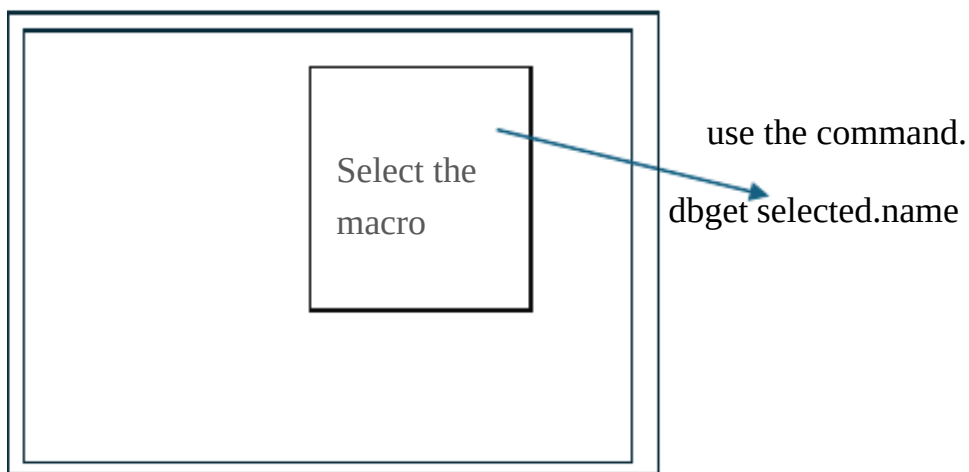
upf commands are made for Synopsys [Multi voltage designs] eg: create_power_domain

CPF: common power format

suppose if we do man create_power_domain in cadence tool? wont-work in cadence. But cadence tool understands upf 1801 IEEE format.

dbget commands : data base information

data base information



BUF*X*

dbget head.libcells.name BUF*X* -innovus command it will give the list of buffers with different drive strengths

get_obj_cells BUF*X* -synopsys dbget top.inst.name #used to report all the instances name present in the design.

dbget selected.pStatus #provide status of selected object, fixed/placed/unplaced.

dbset Command

dbset selected.pStatus fixed #selected object will be fixed.

dbQuery #used to get information about in particular location how many std cells are there

edit commands:

editPowerVia #add via at particular location

editPin #used to place pin/port

INNOVUS

innovus #command used to invoke innovus shell

win #to get gui window

win off #to close gui window

exit #to exit shell

saveDesign placement.enc #To save the block or design

Eg : saveDesign cts.enc.dat

restoreDesign placement.enc block_name #To restore the block or design

In innovus select one object, press q provides detail of that. name, status, orientation, hieght, width etc..

dbget selected.pgTerms.name

VSS VDD

dbget selected.pgTerms.net.name

VSS VDD

#term means pin/port

dbGet top.terms.name #command used to get port names.

set_visible_nets -all #command makes all nets visible.

unplaceAllBlock #command place all macros outside the core .

PD flow

import design ***** sanity checks ***** floorplan powerplan ***** placement ***** cts***** routing.

Import design:

set init_verilog ../../netlist.v

set lef_path ../../all_design.lef

set tech_file ../../file.tf

init_design ##the cmd used to read design using tcl variable.

sanity check:

checkDesign -netlist

check_library

check_instance_library_in_views

check_timing

Floorplan

* Gui: floorplan > specify floorplan > give height, width, spacing values and click okay.

Command: floorplan -d {w h left bottom right top} #to create floorplan using command

Command changeFloorplan -coreToLeft-coreToRight-coreToBottom-coreToTop # Use to modify I/O spacing.

command: changeFloorplan ###used to modify io spacing.

command: setObjFPlanBox Module block1 1 2 3 4 5 Module complete design ###used to create rectilinear shape

Module ### if any complex Designs

Group ## if Any Voltage areas or voltage island available

command: defIn defin floorplan.def ##command used to read def file.

sdcc file: #for both innovus and icc2 sdcc commands are same.

create_constraint_mode

create_rc_corner

set_operation_conditions

source mmmc.tcl

viewdefinition.tcl

all the active scenario for both setup n hold.

vtg information

how to save, free, restore

innovus> saveDesign import_design_done.enc innovus> exit

innovus> restoreDesign import_design.enc.dat block_name

cd./DB/import_design.enc.dat/ ###.enc.dat ---- every stage lib/lef/sdcc/viewdefinition/upetc...

innovus> restoreDesign import_design.enc.dat block_name -noTiming #restore design without timing mode, helps to view, analyse design but we can't save it.

innovus> freeDesign ##close current design in shell

Pin or port placement.

Gui ----edit ----pin editor -----select layer, side, spacing between each port

Command method

setPinConstraint #side, layer, spacing, reverse alternate, origin distance etc...

editPin -pin (pin1 pin2 pin3)

-side

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-edge
-start (040)
-end (40 40)
-layerH (METAL4 METAL6} -layerV (METALS METAL7}
-pattern reverse_alterate
-spacing

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checkPinAssignment #to check ports are legal or illigual or unplaced.

illigual ###ports are not on track.

legalizePi-moveFixedPin -keepLayer -pin (pin1 pin2) #command legalalize the pin

selectIOPin

dbset selected.pStatus fixed

floorplan.def shape, die, core, io space, port placement.

dbget top.terms.name #command reports all the terms/port/pin name.

(pin1 pie2.....}

join [dbget top.terms.name] \n #reports all the terms one after the next.

pin1

pin2

pin3

get_object_name [join [dbget top.terms.name] \n] > portname.rpt #command used to get all the port name in one file.

###get_cell, get_plin, get_net, get_object_name |

###dbget commands

data base get information

----top

----head

----selected

innovus > dbget top. #press tab

#command used to get pointer to the top cell in the design

fplan hinst inst terms instTerms net wire via

: dbget top.terms. #press tab ex

dbget top.terms.name

llength [dbget top.terms.name]

#count of ports present in the design

dbget top.inst.name

#reports all the std. cells names insti

dbget top.inst.cell.name

#all reference name BUF1

dbget top.inst.cell.baseClass block

dbget top.inst.name #reports all the std. cells names inst1

dbget top.inst.cell.name #all reference name BUF1

dbget [dbget top.inst.cell.baseClass -p1 block].name

-p pinter to same location

-p1 pointer to one step back

-p2 -- pointer to two step back

-u uniqfy

dbget top.terms.layer.name -u METAL6 METAL7 METAL8

dbget top.inst.instterms.name

innvovus > dbget head. ##point to the root of the cell

get_lib_cell "BUF #in icc2

dbget head.libcells.name *BUF*LVT

innovus > dbget selected.

#to get fro selected, select any object in ttol.

dbget selected.name

dbget selected.layer.name

dbget selected.layer.net.name

Floorplan area details

* dbget top.fplan.area #to get total area of the block

* dbget top.fplan.box #to get [urx ury llx lly]

* dbget top.fplan.bboxes #to get rectilinear boxes

* dbget top.fplan.box_size #to get width

* dbget top.fplan.box_sizey #to get height

Macro Placement

hier1/hie/cell

hier2/hie/cell2

hier3/hie/cell3

selectInst hier1/hie/cell #14 #command used to select macro cells belong to same hier in gui.

highlight -index 1

#command used to provide color

selectInst hier2/hie/cell #20 #command used to select macro cells belong to same hier in gui.

highlight -index 2 #command used to provide color

placeInstance macro1 (0 0 1 2) #command used to place cell at particular location

dbget selected.orient

checkPlace #to see overlaps.

dbget top.pds.name #power domain name

* free_power_intent #remove if any power domain present

* read_power_intent -1801 upf.upf #read power domain

* commit_power_intent #commit into design

dbget top.pds.name

#report power domains name

dbget top.pds.pgTerms.name

dbget top.pds.pgNet.name

dbget top.pds.density

writeFPlanScript section Group -filename va_details.tcl # to write the script

writeFPlanSscript section blocks -filename macro.tcl

command: report PowerDomain # different power domains in the design information

innovus> help *power* #different power domains commands you can see using these command

How to create row, delete row, cut row_____

deleteRow -all

createRow -site H112-area 1 2 3 4

cutRow

###why we do cut row over macro to avoid, because macro power routing starts from M3, M4, based on requirement. if hard blockage creates, row we have to delete in that region.

selectInst *a/b/c/d * # what are cell present in the hierarchy. And used to highlight the cells in the design

selectInst name1/name2*

press Z for zoom in , shift + z for zoom out

innovus> selectInstByCellName BUF # to highlight reference name of particular cell in the design

innovus> llength [dbget selected.name]

innovus> 2000

innovus> deselectAll

innovus> selectInstByCellName INV

llength [dbget selected.name] #3000

innovus> 3000

port name clk_pll

selectInst clk_pll

selectInstByCellName clk_pll #if you don't know whether it is a pin or port (object not found

selectIOPin clk_pll

selectNet clk_pll_net

selectPin cell/A

select_obj name #if name of the object is pin/port/net/cell/inst -- then use this cmd

To create blockage in the design.

command: createPlaceBlockage -box-type -percentage hard/soft/partial

partial 60% in icc2 60% given, blocked percentage

partial 60% in innovus 60% given, allowed percentage

soft 60% -- in innovus, allow 60% buffer n inverters. # in innovus soft blockage will also come with some percentage

command: deletePlace Blockage name deletePlaceBlockage -all

command: createRouteBlk name -box -layer deleteRouteBlk -all

halo/ keepout margin.

command : addHaloToBlock 1234-cell [dbget top.inst.cell.baseclass -p block]

: deleteHaloFromBlock -all

- ▣ addEndCap #to add endcap cells in the design
- ▣ verifyEndCap #to verify its placement setInstancePlacementStatus -name endcap status fixed #fix cells before next stage.
- ▣ addWellTap -checkerboard -name
- ▣ verifyWellTap

Sanity Checks commands

- ▣ checkFPlan
- ▣ checkPlace
- ▣ checkDesign

Powerplan PG drc violations

clearDrc #to clear existing viol, in d violation browser.

- ▣ Verify_pg_short #to check any power, ground shorts present
- ▣ Verify_drc #to see drc
- ▣ verifyConnectivity #to check any opens
- ▣ editPowerVia #command used to add via
- ▣ Gui: rerouting:

Bind key

* Shift + R --- move/reshape/resize

* C-copy

* k - Ruler

* Shift + k --- delete ruler

* Shift + Y to create placement blockage

* Shift + B --- create routing blockage

* Shift + A --- edit wire

* Shift + X - cut wire by line

F2 window will come, save design

* Z --- zoom in

Shift + Z- zoom out

* f - fit

g-group #select cell n press g

* Shift + g – ungroup

Edit commands

* editSelect -net VDD

* editChangeNet -net VDD2

* editSelect -net name

* editChangeNet-layer_horizontal metal3 -layer_vertical metal4

* editSelect-layer METAL3

* editChangeStatus -to FIXED/ROUTED etc

Unplace Commands

unplaceAllBlocks

dbset [top.inst.cell.baseclass blocks -p].pstatus unplaced

* unplaceAllGuides #this command deletes all the bounds

* unplaceAllInsts I #unplace all standard cell.

innovus> help *edit*

innovus> help *unplace*

innovus> help *verify*

Placement

- region -- hard bound #same hier cell should sit, other cells are allowed.
- guide- soft bound #same hier cell can sit, other cells also allowed.
- fence exclusive bound #same hier cell should sit, other cells are not allowed.

deleteInstGroup name1_region

createInstGroup name1_region

addInstToInstGroup name1_region hier1/hier2/hier3/*

createRegion name1_region 1000 2000 1000 2000

select region In gui

dbget selected.members.name

dbget selected.density

dbget selected.name

Note: Region we can see only in floorplan view, we can't see in placement view. Region's density better to set between 30 to 50.

Different views are available.

- Floorplan view
- Placement view

* setPlaceMode or set_place_mode

* Place_opt_design

Place_design [global place]

opt_design [detailed placement with timing optimization]

* reportCongestion

* report_timing-check_type setup

* checkPlace

* refine Place

place_design [places all the cells based on hierarchy, timing, congestion, without fixing overlaps] optDesign - preCTS [timing optimization, power optimization, drv optimization, area optimization etc...]

place_opt_design

checks after placements

1. checkPlace

overlaps/unplaced cells/ site is not correct / pin track alignments, base drc etc....

reason: improper netlist, less area, improper optimization

solution:

refinePlace

check with synthesis team density screen [partial blk]

cell padding/inst padding

module splitting [setPlaceMode -place_global_module_padding hire1/hire2/hie3*]

Placement gui

- * Place > place standard cell
- * Place > place spare cell
- * Place > specify > cell padding
- * Place > physical cell [end cap, tap, filler]
- * Place > refine placement /
- * Place > scan chain /
- * Place> tie hi tie low /
- * Place > ECO placement

Fixes

Gui: in view window - overlay [enable congestion hotspot/density in gui]

Congestion-density screen

Cell padding - specifyCellPad-cell-left 2-right

deleteCellPad *

o Inst padding -specifyInstPad name -left-right-top-bottom

deleteInstPad -all

* refinePlace

Check utilization

Check timing/fanout

Clock tree synthesis

Gui:

* setOptMode

* Set_ccopt_property

* setPlaceMode

* ccopt_design

* Report_timing-delay_told hold -max_paths 200 >

before_opt_hold_fix

* optDesign -postCTS -hold

* Report_timing-delay_told hold-max_paths 200 > after_opt_hold_fix

Clock tree reports

- Report_clock_tree_structure # clock tree structure buffers name
- Report_ccopt_clock_trees # summary of the clock tree about latency, skew
- Report_ccopt_skew_groups
- timeDesign-postCTS -outDir filename
- timeDesign-hold-postCTS -outDir filename
- Report_clock_timing
- Report_power or reportPower
- Report_area or reportArea

Upsize / down size, add buffer

* ecoChangeCell-inst CELL1/CELL2-cell BUF2X]

[before BUF1X --- BUF2X] - above command used for both, upsize/downsize, vt swap.

* ecoAddRepeater name -inst BUF -loc

* Any time while doing above commands need to setEcoMode, otherwise timing will update.

innovus> setECOMode -help

setECOMode -donttouch true-dontsize true -LECCheck false -update_timing false

ecoAddifepeater

setEcoMode -reset

Routing

* setNanoRoute Mode #signal routing timing driven , metal layer selection

* routeDesign

* optDesign-postRoute # fix the violation and any optimization issues will be fixed

Checks after routing

* Verify_drc

* Verify_Drc-check_short_only

* verifyConnectivity #opens

* ecoRoute

* ecoRoute -fix_drc

Physical cell commands

* addDecap-totCap 100-cell name_decap

* deleteDecap

* addFiller-cell fill-prefix

* deleteFiller

* addTieHiLo -cell tie -prefix

Gui PV related

Violation browser > select violations one by one, shows marker in guy

Verify > verify drc

Verify > verify connectivity

r-select object press r for move/resize/reshape

Shift + x for chopping

Shift + a - to edit wire, drw metal layer

Move the cell Moya

Auto command > ecoRoute -fix drc

#filler, metal filler run PV in separate tool calibre /ICV

Pv analysis and fix

verify_drc

innovus> loadViolationReport -filename drc.out-type Calibre #tool creates violation markers.

verify_drc-check_short_only